



PS1 : AI Powered Trust & Risk Intelligence for Informal Lending

1. Problem Statement

Build an AI-powered lending intelligence platform that enables individuals, families, and communities to make safer and more informed lending decisions. The platform should leverage identity verification, financial document analysis, behavioral insights, trust networks, and explainable AI to evaluate borrower credibility, assess lending risk, detect fraudulent activities, and recommend personalized loan terms. Rather than simply approving or rejecting a loan application, the system should provide transparent, evidence-backed recommendations that help lenders understand borrower trustworthiness, key risk factors, optimal lending terms, and the confidence behind every decision. Every recommendation should be explainable, auditable, and supported by verifiable evidence instead of relying on a black-box AI model.

2. Overview & Background

Informal lending among friends, families, local businesses, and communities remains one of the most common yet underserved forms of finance. Unlike traditional financial institutions, these lending decisions are often driven by personal relationships, incomplete information, and subjective judgment, leading to repayment defaults, fraud, disputes, and broken trust.

While conventional credit scoring systems exclude many borrowers with limited financial history, existing informal lending solutions largely focus on loan tracking instead of helping lenders make informed decisions before issuing a loan.

This challenge encourages participants to build an AI-powered decision support platform that combines borrower verification, dynamic trust scoring, fraud detection, risk prediction, and explainable AI into a unified lending intelligence system. The goal is not to replace human judgment, but to augment it with transparent insights, confidence scores, and evidence-backed recommendations that foster trust between lenders and borrowers.

3. Expected Deliverables

- Intelligent Verification & Trust Profiling
- AI-Powered Risk & Fraud Intelligence
- Personalized Lending Decision Engine
- Explainable Lending Intelligence Dashboard



PS2 : AI Powered Research Reproducibility & Experiment Validation Platform

1. Problem Statement

Build an AI-powered platform that evaluates the reproducibility of machine learning research by bridging the gap between **research papers, source code, experimental configurations, and execution results**. Given a research paper and its associated code repository, the system should automatically extract experimental details, execute reproducible experiments, compare obtained results against published findings, and generate an explainable, evidence-backed reproducibility assessment.

Rather than simply checking if code exists or summarizing a paper, the platform should answer:

- i. **Can the reported experiments actually be reproduced?**
- ii. **Which experimental claims were successfully validated?**
- iii. **Where do reproduced results differ from published results?**
- iv. **What missing information or implementation differences caused these discrepancies?**

Every assessment should be transparent, traceable, and supported by evidence rather than functioning as a black-box AI evaluation.

2. Overview & Background

Reproducibility is fundamental to scientific research, yet reproducing machine learning papers remains a significant challenge. Critical experimental details are often scattered across research papers, supplementary material, source code, configuration files, datasets, and documentation. Small differences in preprocessing, hyperparameters, software environments, dataset versions, or implementation choices can lead to substantially different results.

While research repositories have become increasingly accessible, there is currently no unified system that can automatically understand a paper, connect it to its implementation, execute experiments, and objectively evaluate whether published findings can be reproduced.

This challenge invites participants to build an intelligent research validation platform that automates the end-to-end reproducibility workflow—from extracting experimental specifications and running experiments to analyzing discrepancies and generating explainable reproducibility reports. The objective is not to replace researchers, but to accelerate scientific validation by providing transparent, evidence-driven insights into the reproducibility of published work.

For the hackathon, organizers will provide a curated collection of lightweight ML papers and corresponding repositories whose experiments can be executed on CPU-only environments.

3. Expected Deliverables

- Automated Research Paper & Repository Analysis
- Experiment Reproduction & Result Validation
- Evidence-Based Reproducibility Assessment
- Interactive Reproducibility Dashboard & Insights



PS3: AI-Powered Court Document Digitization and Multilingual Litigant Assistant

Background:

Indian courts handle over 40 million pending cases, and a large share of case records, from affidavits and daily orders to judgments, still exist as scanned paper documents or handwritten filings that are neither searchable nor easily shared across case management systems. Legal staff and litigants routinely struggle to extract key information from lengthy documents, and citizens who are not fluent in English or legal terminology often have no accessible way to understand the status or content of their own case. Platforms such as Adalat AI have shown that combining document intelligence with multilingual chat access can meaningfully reduce this gap, using AI powered OCR to convert scanned records into structured, searchable text and WhatsApp based chatbots to deliver case updates and summaries in regional languages. Most courts and legal aid organizations, however, still lack any accessible version of this pipeline, relying on manual data entry for digitization and leaving litigants dependent on lawyers or clerks for basic case information.

Problem Statement:

Design and build a web platform that digitizes scanned legal documents into structured, searchable text, generates concise summaries of lengthy case documents, and allows users to query case information through a multilingual chat interface, making court records more accessible to both legal staff and everyday citizens.

Expected Deliverables:

- A document upload and OCR pipeline that converts scanned or photographed legal documents into structured, searchable text using an existing OCR service or model
- A summarization module that condenses long legal documents such as judgments, affidavits, or case notes into short, plain language summaries
- A translation layer that renders summaries and key case details in at least one Indian regional language alongside English
- A chat interface where a user can ask questions about an uploaded case document, such as case status, key parties, or next hearing date, and receive answers grounded in the actual document content
- A simple case dashboard listing uploaded documents with basic metadata such as case number, document type, and upload date, allowing quick search and retrieval



PS4: Personalized Voice AI Companion

Background:

Voice AI assistants, from customer support bots to companion apps and smart hardware devices, are increasingly deployed for repeated, ongoing interactions with the same user, yet most systems treat each conversation as stateless or rely on replaying full transcripts, which is token expensive and prone to losing important context such as a user's preferences, recurring topics, and emotional patterns buried in long histories. Recent research such as VoiceMem, published in 2025, demonstrates that structuring conversational memory as a dual-layer graph, with factual entities and knowledge on one side and personality and emotional context on the other, allows a voice agent to recall relevant user context using as few as five targeted memories instead of replaying dozens of past turns, while cutting response latency by an order of magnitude compared to baseline retrieval approaches. At the same time, personalization increasingly depends on more than conversation history alone. A user's music taste, listening habits, and social media activity carry rich, continuously updating signals about mood, interests, and identity that most voice assistants never tap into, leaving companion apps feeling generic rather than genuinely personal. Small teams building voice or chat companions today either forget everything between sessions or bolt on naive vector search over transcripts, missing both the structured-memory advantage and the personalization available from a user's broader digital footprint.

Problem Statement:

Design and build a voice or chat AI companion with a persistent memory system that extracts and structures facts and preference signals from conversations, enriches that memory using connected music and social media activity, and retrieves only the most relevant memories to generate context aware, personalized responses, while demonstrably reducing context size and improving relevance compared to naive full transcript retrieval.

What can be built:

- Memory Extraction & Structuring Pipeline
- Structured Memory Store & Knowledge Graph
- Context-Aware Memory Retrieval & Ranking
- Music & Social Context Integration
- Persistent Chat/Voice Companion Interface
- Memory Performance & Relevance Evaluation



PS5: AI-Powered Deepfake and Synthetic Media Detection Platform

Background:

The rapid proliferation of generative AI tools has made photorealistic image, video, and voice synthesis accessible to anyone, with deepfake incidents in India rising sharply : the *Ministry of Electronics and IT* (MeitY) reported a marked increase in deepfake-related complaints through 2025, spanning political disinformation, non-consensual synthetic imagery, and financial fraud (including voice-cloned "family emergency" scam calls). Most existing detection tools are either proprietary enterprise products (*Reality Defender, Hive Moderation*) inaccessible to individual journalists, fact-checkers, and ordinary citizens, or academic research models with no usable interface. Meanwhile platforms and users have no lightweight, self-serve way to check whether an image, video clip, or voice note they've received is likely synthetic before sharing or acting on it, leaving verification to slow, manual forensic analysis that most people cannot perform.

Problem Statement:

Design and build a web platform that accepts an uploaded image, short video, or audio clip and returns a confidence score on whether it is AI-generated or manipulated, using a trained or fine-tuned classification model, with a clear, explainable presentation of the signals driving that verdict rather than a single opaque number.

Expected Deliverables:

- AI-Powered Media Authenticity Detection
- Interactive Media Analysis Interface
- Explainable Detection & Visual Evidence
- Batch Analysis & Detection History
- Synthetic Audio Detection